Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/24/2015
PI Name:	Mednieks, Maija Ph.D.		
Project Title:	Protein Expression in Salivary Glands: Effects of Extended Space Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	mednieks@nso1.uchc.edu , mednieks@uchc.edu	**Fax:**	FY 860-679-1342
PI Organization Type:	UNIVERSITY	**Phone:**	860-679-4140 4140
Organization Name:	University of Connecticut Health Center		
PI Address 1:	Department of Oral Health and Diagnostic Sciences		
PI Address 2:	263 Farmington Ave.		
PI Web Page:			
City:	Farmington	**State:**	CT
Zip Code:	06030-0001	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	06/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; STS-133 ; Bion M1 NOTE: Extended to 6/30/2015 (from 12/31/2014), per NSSC information (Ed., 12/16/14) NOTE: Extended to 12/31/2014, per A. Chu/ARC and NSSC information (Ed., 6/12/14) NOTE: Extended to 7/31/14 per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hand, Arthur D.D.S. (University of Connecticut Health Center)		
Grant/Contract No.:	NNX09AP13G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The study was designed to determine effects of extended space travel on oral tissues using the salivary glands of mice. Rodent salivary glands are structurally and functionally similar to those of humans therefore useful for research on changes associated with travel in space. Salivary glands and oral fluid play an important role in maintaining the integrity of the mouth and teeth. Proteins present in secretion granules of salivary glands are released into saliva and can be used to monitor effects of environmental stress. Salivary proteins can serve as markers for physiologic processes such as digestive and immune systems and hormonal regulation. The proteins in saliva are reliable indicators of the environmental effects of travel in space. The physiologic processes for which they are markers are either slowed, accelerated or unaffected. Several of the proteins have previously been shown to respond to a variety of environmental stimuli, including space flight. Our previous studies on short duration (~ 1 week) space flight missions showed that specific secretory proteins were modified by exposure to both zero (ØG) and hyper G conditions. Proteins associated with hormonal signaling and gene expression are down-regulated. An enzyme that is associated with digestion seems unaffected while some cellular proteins appear increased. Employing saliva as a diagnostic test medium could have clinical applications and be useful in measuring human responses to space flight. The present results agree with previous findings that proteins in salivary glands are specifically altered during space flight and can therefore be used as markers for environmental effects on different physiological and behavioral responses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Background and Objectives: It is important to study the effects of travel in space and to measure reactions to the stress of weightlessness in zero gravity (ØG) conditions. Stress changes hormone action that is reflected in glands that produce saliva as well as in different body parts/organs. The goal is to devise a clinical test for stress disorders in space and on Earth. Saliva contains substances (e.g. proteins) that change during travel in space. We can determine which salivary proteins have been changed by using biochemical tests and histologic analyses to study tissues and cells (salivary glands) that produce saliva in mice flown on the space shuttle Discovery STS-131, 133, and 135 missions. This information will be used to design a test kit to measure responses in astronauts so no blood will have to be drawn. Collecting saliva instead of blood and urine, will be easier during space flight and will make it possible to study hormone changes experienced by Astronauts and Cosmonauts at ØG. Travel in space is an ideal model system for studying a specific environmental impact on physiologic function that is due to gravitational change and cannot be duplicated on Earth. Understanding the effect of ØG will define changes due to the environment as the stressor. This information can then be adapted to devising specific tests of different conditions using saliva for general clinical applications in medicine in space or on Earth. Research Description: The Biospecimen Sharing Program has provided an unparalleled opportunity to study tissue and secretory protein signaling in tissues affected by environmental stimuli whereby cellular and molecular changes associated with travel in space can be identified in experimental animals and can be monitored by measuring the expression of specific proteins. Previous studies on the effects of space flight have shown that regulation by catecholamine hormones is markedly altered during space flight. The expression of specific proteins as markers can potentially be used for measuring responses to environmental stress since changes in the expression of secretory proteins are reflected in saliva.
Task Progress:	The present results indicate that travel in space alters the expression of specific proteins. Notably, proteins associated with hormonal signaling and gene expression in salivary glands are initially down-regulated, other secretory proteins appear increased, and still others are unaffected. The observed phenomena are therefore related to function rather than being due to some overall, nonspecific physiologic functional disruption. The responses to microgravity by secretory proteins that may be specific for metabolic, signaling pathways or be indicatory of digestive or immune processes can be the basis for designing measuring devices to test for the disruption of these processes in astronauts. Ultimately such tests can be adapted for clinical or field use on Earth.
Bibliography Type:	Description: (Last Updated: 02/04/2021)
Abstracts for Journals and Proceedings	Mednieks MI, Hand AR. "Microgravity-associated changes in cellular signal processing." Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. Committee on Space Research (COSPAR) 2014 40th Scientific Assembly, Moscow, Russia, August 2-10, 2014. , Aug-2014
Abstracts for Journals and Proceedings	Mednieks MI, Haxhi C, Hand AR. "Secretory Protein Expression in Oral Tissues After BION M-1 and US Space Shuttle Missions." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Articles in Peer-reviewed Journals	Mednieks M, Khatri A, Rubenstein R, Burleson JA, Hand AR. "Microgravity alters the expression of salivary proteins." Oral Health Dent Manag. 2014 Jun;13(2):211-6. PubMed PMID: 24984624 , Jun-2014
Articles in Peer-reviewed Journals	Zinn VZ, Khatri A, Mednieks MI, Hand AR. "Localization of the PKA regulated CFTR-PKA-Ezrin/Cl- channel Signaling Complexes in Striated Duct Cells of Human Salivary Glands." Arch Oral Biol. (Under review, as of January 2015). , Jan-2015